CrossSum: Beyond English-Centric Cross-Lingual Summarization for 1,500+ Language Pairs

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Abstract

We present CrossSum, a large-scale cross-lingual summarization dataset comprising 1.68 million article-summary samples in 1,500+ language pairs. We create CrossSum by aligning parallel articles written in different languages via cross-lingual retrieval from a multilingual abstractive summarization dataset and perform a controlled human evaluation to validate its quality. We propose a multistage data sampling algorithm to effectively train a cross-lingual summarization model capable of summarizing an article in any target language. We also introduce LaSE, an embedding-based metric for automatically evaluating model-generated summaries. LaSE is strongly correlated with ROUGE and, unlike ROUGE, can be reliably measured even in the absence of references in the target language. Performance on ROUGE and LaSE indicate that our proposed model consistently outperforms baseline models. To the best of our knowledge, CrossSum is the largest cross-lingual summarization dataset and the first ever that is not centered around English. We are releasing the dataset, training and evaluation scripts, and models to spur future research on cross-lingual summarization. The resources can be found at https://github.com/csebuetnlp/CrossSum.

1 Introduction

Cross-lingual summarization (hereinafter XLS) is the task of generating a summary in a target language given a source text in another language. The task is challenging as it combines summarization and translation in one task, both challenging tasks in their own right. Earlier approaches to XLS thus employed pipeline methods such as translate-then-summarize (Leuski et al., 2003) and summarize-then-translate (Wan et al., 2010). Not only are they computationally expensive, having to use multiple models, but these approaches also suffer from error-propagation (Zhu et al., 2019) from one model to another, degrading the overall performance.

Input Article: [...] 新型コロナウイルスに対し、様々な既存の治療法の効果を試す世界的規模の臨床試験の一貫として、デキサメタゾンが試された。(Dexamethasone was tested as part of a global clinical trial to test the effectiveness of various existing therapies against the new coronavirus.) [...] その結果、人工呼吸器を必要とする重症患者の致死率が3割下がり。(As a result, the case fatality rate of critically ill patients who require a ventilator is reduced by 30%.) [...] ボリス・ジョンソン英首相は「イギリス科学界の素晴らしい成果」を歓迎し。(British Prime Minister Boris Johnson welcomed "the great achievements of the British scientific community".) [...] 「しかもこれは、世界中で手に入る薬だ」("And this is a medicine available all over the world".) [...] きわめて安いステロイド剤だった (but a very cheap steroid that has been used for a long time.)

Summary: বিজ্ঞানীরা বলছেন ডেক্সামেথাসোন নামে সস্তা ও সহজলভ্য একটি ওষুধ করোনাভাইরাসে গুরুতর অসুস্থ রোগীদের জীবন রক্ষা করতে সাহায্য করবে। (Scientists say a cheap and readily available drug called dexamethasone will help save the lives of critically ill patients with coronavirus.)

Figure 1: A sample article-summary pair from CrossSum, the article is written in Japanese, and the summary is in Bengali. We translate the texts to English inside parentheses for better understanding. Words and phrases of the article relevant to the summary are color-coded.

The success of sequence-to-sequence (seq2seq) models (Cho et al., 2014; Sutskever et al., 2014) and the advances in Transformer-based models (Vaswani et al., 2017) have aided in the emergence of end-to-end methods that can perform XLS with one single model (Zhu et al., 2019; Cao et al., 2020b). The availability of XLS datasets (Ladhak et al., 2020; Perez-Beltrachini and Lapata, 2021) has also helped this task gain popularity in recent times. However, they cover only a few languages, contain a small number of samples for training and evaluation, or use English as the pivot language (i.e., the target language always remains English), thereby limiting their applicability to a great extent.

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To democratize XLS beyond high-resource languages, in this work, we introduce **CrossSum**, a large-scale XLS dataset containing 1.68 million article-summary samples in 1,500+ language pairs. We align parallel articles[1] written in different languages via cross-lingual retrieval from the multilingual XL-Sum (Hasan et al., 2021) dataset. We introduce and rigorously study the notions '*induced pairs*' and '*implicit leakage*' to increase the coverage of the dataset while at the same time ensuring maximum quality. We also perform a controlled human evaluation of CrossSum spanning nine languages from high- to low-resource and show that the alignments are highly accurate.

We design **MLS**, a multistage language sampling algorithm, for successfully training models that can generate a summary in any target language for an input article in any source language, both from a set of languages present in the training dataset. For the first time, we perform XLS with CrossSum on a broad and diverse set of languages without relying on English as the standalone pivot, consistently outperforming many-to-one and one-to-many models, as well as summarize-then-translate baselines.

We propose **LaSE**, an embedding-based metric for evaluating summaries when reference summaries may not be available in the target language but may be available in another language, potentially opening new doors for evaluating low-resource languages. Furthermore, we demonstrate the reliability of LaSE by its high correlation with ROUGE (Lin, 2004), the de-facto metric for evaluating text summarization systems.

To the best of our knowledge, CrossSum is the largest publicly available abdtractive XLS dataset, both in terms of the number of samples and the number of language pairs. We are releasing the dataset, training and evaluation scripts, and models hoping that these resources will encourage the community to push the boundaries of XLS beyond English and other high-resource languages.

2 The CrossSum Dataset

The most straightforward way of curating a high-quality XLS dataset is via crowd-sourcing (Nguyen and Daumé III, 2019). However, it may be difficult to find crowd workers having professional command over low-resource languages or distant language pairs. Moreover, scalability issues might arise due to the time and budget constraints for crowd-sourcing. Therefore, synthetic (Zhu et al., 2019) and automatic methods (Ladhak et al., 2020; Perez-Beltrachini and Lapata, 2021) have gained traction over crowd-sourcing.

Automatic curation of an XLS dataset is simply to pair an article `A` in a source language with the summary of a parallel article `B` written in a different target language (Figure 1), assuming the availability of a multilingual dataset having identical contents in different languages. Two contemporary works have compiled large-scale multilingual summarization datasets, namely XL-Sum (Hasan et al., 2021) (1.35M samples in 45 languages) and MassiveSumm (Varab and Schluter, 2021) (28.8M samples in 92 languages). Though substantially larger than the other, MassiveSumm is not publicly available. Since public availability is crucial for promoting open research, we opted for XL-Sum, distributed under a non-commercial license. Additionally, all articles of XL-Sum are crawled from a single source, BBC News. We observed that BBC publishes similar news content in different languages and follow similar summarization strategies. Hence adopting XL-Sum would increase the quality and quantity of the article-summary pairs.

Unlike previous automatic methods, there are no explicit links between parallel articles in XL-Sum. Fortunately, language-agnostic sentence representations (Artetxe and Schwenk, 2019a; Feng et al., 2022) have achieved state-of-the-art results in cross-lingual text mining (Artetxe and Schwenk, 2019b), and hence, we use them to search identical contents across languages. For simplicity[2], we perform the search over summaries only. To ensure maximum quality, we set two conditions for a summary S_A in language `A` to be aligned with another summary S_B in language `B`:

1. S_B must be the nearest neighbor of S_A among all summaries in `B`, and vice-versa.
2. The similarity between S_A and S_B must be above the threshold, τ.

The similarity of a summary pair is measured by the inner product of their Language-agnostic BERT Sentence Embeddings (LaBSE) (Feng et al., 2022) (a unit vector for an input text sequence). We empirically set the similarity threshold as the average over all languages that maximized their respective F_1 score ($\tau = 0.7437$) in the BUCC mining tasks (Zweigenbaum et al., 2017).[3]

[1] We re-purpose the terminology of parallel corpus here.

[2] The entire procedure is described in Appendix A.

[3] Around 90% F_1 is achieved using LaBSE in BUCC, hence not all CrossSum alignments will be correct. Therefore,

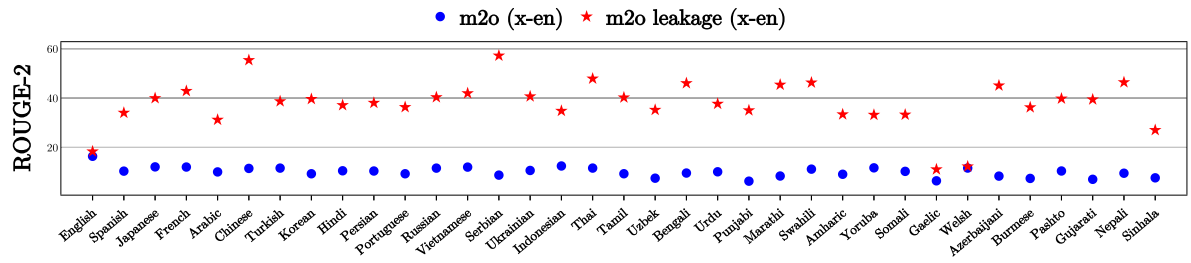


Figure 2: Training on the dataset respecting the original XL-Sum splits causes unusually high ROUGE scores (marked red) in many-to-one models due to implicit data leakage. Therefore, we redid the splits taking the issue into account, and consequently, models trained on the new set (marked blue) do not exhibit any unusual spike.

Induced Pairs We observed that many summary pairs, despite being nearest neighbors in their language pairs, were filtered out because of the threshold τ. Although interestingly, both were aligned with the same summary in a different language. Moreover, these pairs are prevalent if their languages are distant or low-resource. LaBSE uses contrastive learning (Guo et al., 2018; Yang et al., 2019) to rank parallel sentences over non-parallels. Since parallel pairs are mostly found for high-resource and linguistically close languages, we hypothesize that LaBSE fails to assign high similarity to sentences from languages that are not.

To include these pairs into CrossSum, we introduce the notion '*induced pairs*.' Formally, two summaries S_A, S_B in languages A, B are induced pairs if they are nearest neighbors of each other in A, B, their similarity score is below τ, and both are aligned with S_C in language C, or through a chain of aligned pairs $(S_A, S_C), (S_C, S_D), \cdots, (S_Y, S_Z), (S_Z, S_B)$ in languages $\{\mathtt{C}, \mathtt{D}, \cdots, \mathtt{Y}, \mathtt{Z}\}$.

We thus incorporate the induced pairs into CrossSum through a simple graph-based algorithm. First, we represent all summaries as vertices in a graph and draw an edge between two vertices if the summaries are aligned. Then we find the connected components in the graph and draw edges (i.e., induced pairs) between all vertices in a component. Again to ensure quality, before computing the induced pairs, we use the max-flow min-cut theorem (Dantzig and Fulkerson, 1955) considering the similarity scores as edge weights to limit the size of each component to 50 vertices (since ideally, a component should have at most 45 vertices, one summary from each language) and set their minimum acceptance threshold to $\tau' \leftarrow \tau - 0.10$.

in the following section, we further assess the quality of the alignments using human evaluation.

We finally assembled the originally aligned pairs and induced pairs to create the CrossSum dataset. Figure 6 (Appendix) shows the article-summary statistics for all language pairs in CrossSum. As evident from the figure, CrossSum is not centered only around the English language but rather distributed across multiple languages.

Implicit Leakage We initially made the train-dev-test splits respecting the original XL-Sum splits and performed an initial assessment of CrossSum by training a many-to-one model (articles written in any source language being summarized into one target language). Upon evaluation, we found very high ROUGE-2 scores (around 40) for many language pairs, even reaching as high as 60 for some (Figure 2). In contrast, Hasan et al. (2021) reported ROUGE-2 in the 10-20 range for the multilingual summarization task.

We inspected the model outputs and found that many summaries were the same as the references. Through closer inspection, we found that their corresponding articles had a parallel counterpart occurring in the training set in some other language. During training, the model was able to align the representations of parallel articles (albeit written in different languages) and generate the same output by memorizing from the training sample. While models should undoubtedly be credited for being able to make these cross-lingual mappings, this is not ideal for benchmarking purposes as this creates unusually high ROUGE scores. We denote this phenomenon as '*implicit leakage*' and make a new dataset split to avoid this. Before proceeding, we deduplicate the XL-Sum dataset[4] using semantic similarity, considering two summaries S_A, S'_A in language A to be duplicates of one another if

[4]XL-Sum has been deduplicated using lexical overlap methods only. But due to the risk of implicit leakage, which is not lexical, we further perform semantic deduplication.

their LaBSE representations have similarity above 0.95. We take advantage of the component graph mentioned previously to address the leakage and assign all article-summary pairs originating from a single component in the training (dev/test) set of CrossSum, creating an 80%-10%-10% split for all language pairs. Since parallel articles no longer appear in the training set of one and the dev/test set of another, the leakage is not observed anymore (Figure 2). We further validated this by inspecting the model outputs and found no exact copies.

3 Human Evaluation of CrossSum

To establish the validity of our automatic alignment pipeline, we conducted a human evaluation to study the quality of the cross-lingual alignments.

We selected all possible combinations of language pairs from a list of nine languages ranging from high-resource to low-resource to assess the alignment quality in different pair configurations (e.g., high-high, low-high, low-low) as per the language diversity categorization by Joshi et al. (2020). We chose three high-resource languages, English, Arabic, and (simplified) Chinese (categories 4 and 5); three mid-resource languages, Indonesian, Bengali, and Urdu (category 3); and three low-resource languages, Punjabi, Swahili, and Pashto (categories 1 and 2), as representative languages and randomly sampled fifty cross-lingual summary alignments from each language pair for annotation. As a direct evaluation of these pairs would require bilingually-proficient annotators for both languages, which are practically intractable for distantly related languages (e.g., Bengali-Swahili), we resorted to a pivoting approach during annotation for language pairs that do not contain English. For a language pair $(l_1 - l_2)$, where $l_1 \neq en$ and $l_2 \neq en$, we sampled alignments (x, y) such that $\exists (x, e) \in (l_1 - en)$ and $\exists (y, e) \in (l_2 - en)$, for an English article e. In other words, we ensure that both the articles of the sampled cross-lingual pair have a corresponding cross-lingual pair with an English article. An alignment (x, y) would be deemed correct if both (x, e) and (y, e) are correct. This formulation thus reduced the original problem to annotating samples from language pairs $(l_1 - en)$ and $(l_2 - en)$, where l_1 and l_2 are from the previously selected languages that are not English.

We hired bilingually proficient expert annotators adept in the language of interest and English. Two annotators labeled each language pair where one language is English. We presented them with corresponding summaries of the cross-lingual pairs (and optionally the articles themselves) and elicited yes/no answers to the question:

"Can the provided sequences be considered summaries for the same article?"[5]

We deem a sequence pair accurate if both annotators judge it as valid. We show the alignment accuracies of the language pairs in Figure 3.

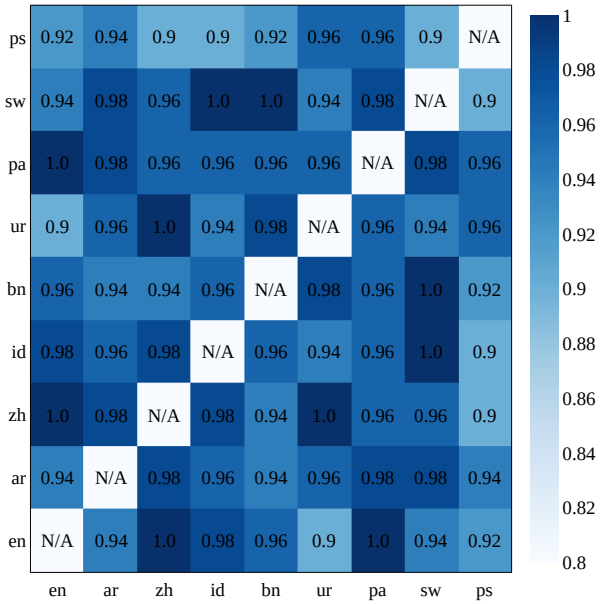


Figure 3: A heatmap showing alignment accuracies of different language pairs obtained by human evaluation.

As evident from the figure, the annotators judge the aligned summaries to be highly accurate, with an average accuracy of 95.67%. We used Cohen's Kappa (Cohen, 1960) to establish the inter-annotator agreement and show the corresponding statistics in Table 3 in the Appendix.

4 Training & Evaluation Methodologies

In this section, we discuss the multistage sampling strategy for training cross-lingual text generation models and our proposed metric for evaluating model-generated summaries.

4.1 Multistage Language Sampling (MLS)

From Figure 6, it can be observed that CrossSum is heavily imbalanced. Thus, training directly without upsampling low-resource languages may result in their degraded performance. Conneau et al. (2020)

[5]We do not explicitly evaluate article-summary correctness as this has already been studied in work on XL-Sum. This was also done to reduce annotation costs.

used probability smoothing for upsampling in multilingual pretraining and sampled all examples of a batch from one language. However, extending this technique to the language pairs in CrossSum would result in many batches having repeated samples as many language pairs do not have enough training samples in total compared to the batch sizes used in practice (e.g., Conneau et al. (2020) used a batch size of 256, which exceeds the training set size of nearly 1,000 language pairs in CrossSum). At the same time, many language pairs would not be sampled during training for lack of enough training steps (due to our constraints on computational resources). To address this, we adapt their method to introduce a **M**ultistage **L**anguage **S**ampling algorithm (**MLS**) to ensure that the target summaries of a batch are sampled from the same language.

Let $L_1, L_2, \ldots, L_n$ be the languages of a cross-lingual source-target dataset, and c_{ij} be the number of training samples where the target is from L_i and source from L_j. We compute the probability p_i of each target language L_i by

$$p_i = \frac{\sum_{k=1}^{n} c_{ik}}{\sum_{j=1}^{n}\sum_{k=1}^{n} c_{jk}} \quad \forall i \in \{1, 2, \ldots, n\}$$

We then use an exponent smoothing factor α and normalize the probabilities

$$q_i = \frac{p_i^{\alpha}}{\sum_{j=1}^{n} p_j^{\alpha}} \quad \forall i \in \{1, 2, \ldots, n\}$$

Given the target language L_i, we now compute the probability of a source language L_j, represented by $p_{j|i}$.

$$p_{j|i} = \frac{c_{ij}}{\sum_{k=1}^{n} c_{ik}} \forall j \in \{1, 2, \ldots, n\}$$

We again smooth $p_{j|i}$ by a factor β and obtain the normalized probabilities

$$q_{j|i} = \frac{p_{j|i}^{\beta}}{\sum_{k=1}^{n} p_{k|i}^{\beta}} \forall j \in \{1, 2, \ldots, n\}$$

Using the probabilities, we describe the training process with the MLS algorithm in Algorithm 1.

Note that the proposed algorithm can be applied to any cross-lingual seq2seq task where both the source and target languages are imbalanced.

Algorithm 1: Multistage Language Sampling (MLS)

Input: $D_{ij}\ \forall i, j \in \{1, 2, \ldots, n\}$: training data with tgt/src languages L_i/L_j;
$c_{ij} \leftarrow |D_{ij}|\ \forall i, j \in \{1, 2, \ldots, n\}$;
m: number of mini-batches.

Compute $q_i, q_{j|i}$ using c_{ij}
while *(`Model Not Converged`)* **do**
  $batch \leftarrow \phi$
  Sample $L_i \sim q_i$
  for $k \leftarrow 1$ **to** m **do**
    Sample $L_j \sim q_{j|i}$
    Create mini-batch mb from D_{ij}
    $batch \leftarrow batch \cup \{mb\}$
  Update model parameters using $batch$

4.2 Evaluating Summaries Across Languages

A sufficient number of reference samples are essential for the reliable evaluation of model-generated summaries. However, for many CrossSum language pairs, even the training sets are small, let alone the test sets (the median size is only 33). For instance, the Japanese-Bengali language pair has 34 test samples only, which is too few for reliable evaluation. But the size of the in-language[6] test sets of Japanese and Bengali are nearly 1,000. Being able to evaluate against reference summaries written in the source language would thus alleviate this insufficiency problem by leveraging the in-language test set of the source language.

For this purpose, cross-lingual similarity metrics that do not rely on lexical overlap (i.e., unlike ROUGE) are required. Embedding-based similarity metrics (Zhang et al., 2020; Zhao et al., 2019) have recently gained popularity. We draw inspiration from them and design a similarity metric that can effectively measure similarity across languages in a language-independent manner. We consider three essential factors:

1. Meaning Similarity: The generated and reference summaries should convey the same meaning irrespective of their languages. Just like our alignment procedure from Section 2, we use LaBSE to compute the meaning similarity between the generated (s_{gen}) and reference summary (s_{ref}):

$$\text{MS}(s_{gen}, s_{ref}) = \text{emb}(s_{gen})^{\text{T}}\text{emb}(s_{ref})$$

where $\text{emb}(s)$ denotes the embedding vector output of LaBSE for input text s.

2. Language Confidence: The metric should identify, with high confidence, that the summary is indeed being generated in the target language. As such, we use the *fastText* language-ID classifier

[6]Both article and summary belonging to the same language

(Joulin et al., 2017) to obtain the language probability distribution of the generated summary and define the Language Confidence (LC) as:

$$\text{LC}(s_{gen}, s_{ref}) = \begin{cases} 1, \text{ if } L_{ref} = \text{argmax}\, P(L_{gen}) \\ P(L_{gen} = L_{ref}), \text{ otherwise} \end{cases}$$

3. Length Penalty: Generated summaries should not be unnecessarily long, and the metric should penalize long summaries. While model-based metrics may indicate how similar a generated summary is to its reference and language, it is unclear how they can be used to determine its brevity. As such, we adapt the BLEU (Papineni et al., 2002) brevity penalty to measure the length penalty:

$$\text{LP}(s_{gen}, s_{ref}) = \begin{cases} 1, \text{ if } |s_{gen}| \leq |s_{ref}| + c \\ \exp(1 - \frac{|s_{gen}|}{|s_{ref}|+c}), \text{ otherwise} \end{cases}$$

s_{gen} and s_{ref} may not be of the same language, and parallel texts may vary in length across languages. Hence, we use a length offset c to avoid penalizing generated summaries slightly longer than the references. By examining the standard deviation of mean summary lengths of the languages, we set $c = 6$.

We finally define our metric, **L**anguage-**a**gnostic **S**ummary **E**valuation (**LaSE**) score as follows.

$$\text{LaSE}(s_{gen}, s_{ref}) = \text{MS}(s_{gen}, s_{ref}) \times \text{LC}(s_{gen}, s_{ref}) \times \text{LP}(s_{gen}, s_{ref})$$

5 Experiments & Discussions

One model capable of generating summaries in any target language for an input article from any source language is highly desirable. However, it may not be the case that such a 'many-to-many' model (m2m in brief) would outperform many-to-one (m2o) or one-to-many (o2m) models[7], which are widely-used practices for XLS (Ladhak et al., 2020; Perez-Beltrachini and Lapata, 2021). In this section, we establish that the m2m model, trained in the presence of samples from all possible language pairs using the MLS algorithm from Section 4, consistently outperforms m2o, o2m, and summarize-then-translate (s.+t.) baselines given equal training steps.

In addition to the proposed m2m model, we train five different m2o and o2m models using five highly spoken[8] and typologically diverse pivot (i.e., the 'one' in m2o and o2m) languages: English, Chinese (simplified), Hindi, Arabic, and Russian. As another baseline, we use a summarize-then-translate pipeline. As fine-tuning pretrained language models (Devlin et al., 2019; Xue et al., 2021a) have shown state-of-the-art results on monolingual and multilingual text summarization (Rothe et al., 2020; Hasan et al., 2021), we fine-tune each model using a pretrained mT5 (Xue et al., 2021a) by providing explicit cross-lingual supervision. We show the results on ROUGE-2 F1 and LaSE in Figures 4 and 5[9]. We limit our evaluation only to the languages supported by mT5, fastText, and M2M-100 (the translation model used in s.+t.).

Results indicate that the m2m model consistently outperforms m2o, o2m, and s.+t., with an average ROUGE-2 (LaSE) score of 8.15 (57.15) over all languages tested, 3.12 (9.02) above s.+t. Moreover, compared to the o2m models on language pairs where the pivots are the targets, the m2m model scores 1.80 (5.84) over m2os, and on those where the pivots are the sources, 6.52 (51.80) over o2ms.

Upon inspection of the model outputs, we found the m2o models to be able to generate non-trivial summaries. In contrast, the o2m models completely failed to produce cross-lingual summaries, performing in-language summarization (the language of the summary is the same as that of its input article) for all targets. We hypothesize that varying the target language in a batch hampers the decoder's ability to generate from a specific language, possibly because of the vast diversity of target languages in the batch (discussed further in Appendix E). s.+t. performed well on high-resource languages but poorly on low-resource ones. This was revealed to be a limitation of the translation model used in the pipeline.

5.1 Zero-shot Cross-lingual Transfer

The previous experiments were done in a fully supervised fashion. However, for many low-resource language pairs, samples are not abundantly available. Hence, it is attractive to be able to perform zero-shot cross-lingual generation (Duan et al., 2019) without relying on any labeled examples.

To this end, we fine-tuned mT5 with only the in-language samples (i.e., the source and target both have the same language) in a multilingual fashion and, during inference, varied the target language. Unfortunately, the model totally fails at generating

[7] Discussed in detail in Appendix C.

[8] https://w.wiki/Pss

[9] A detailed description of the training procedures and hyperparameter choices are detailed in Appendix D.1.

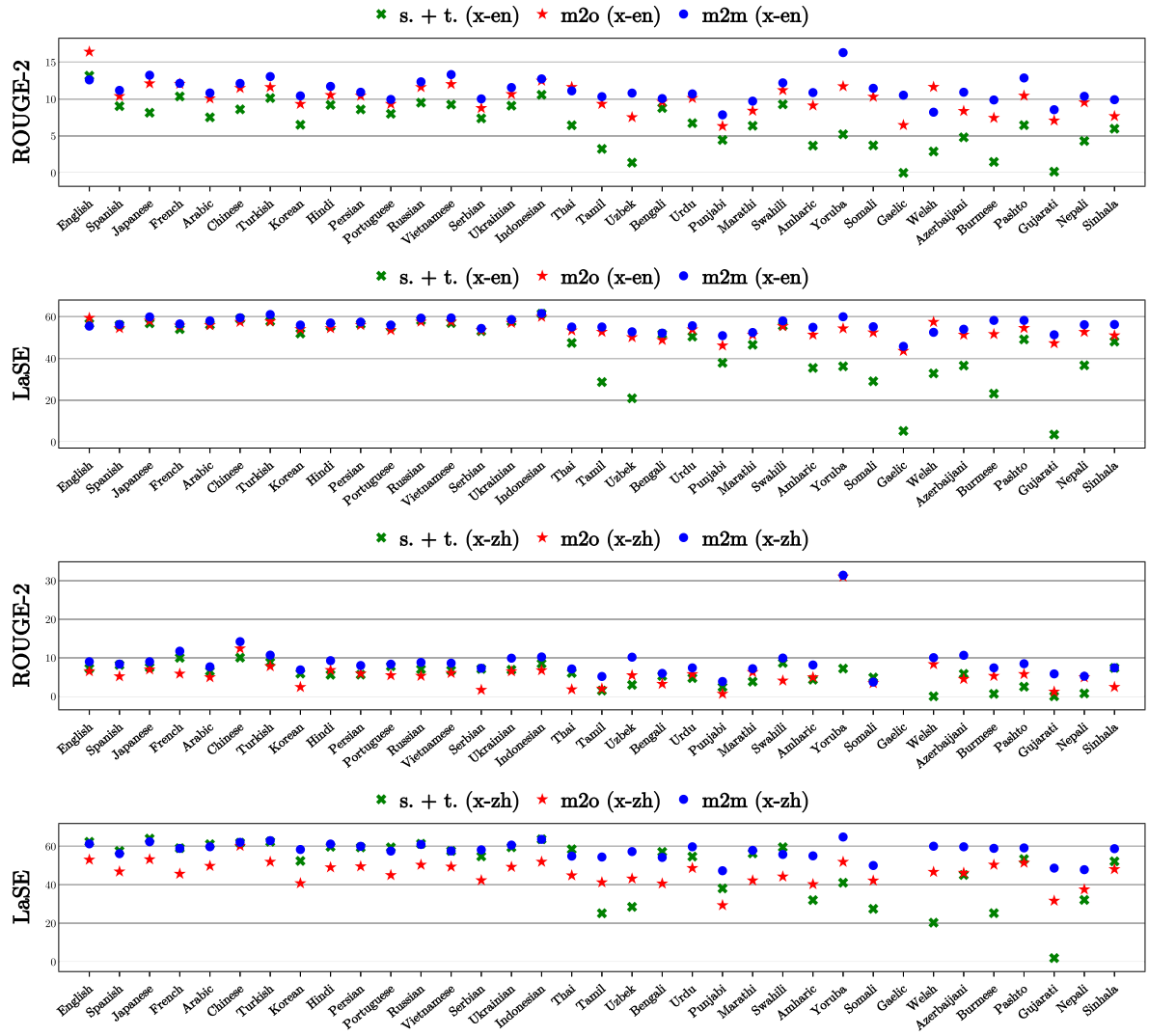


Figure 4: ROUGE-2 and LaSE scores for English and Chinese as target languages as the source languages vary. The m2m model significantly outperforms the m2o models and summarize-then-translate baseline in most languages. The comparisons with other target languages are shown in the Appendix (Figure 8) due to space limitations.

cross-lingual summaries and performs in-language summarization instead.

We also fine-tuned m2o models (with only the in-language samples of the target language) in a monolingual fashion and ran inference in a zero-shot setting with samples from other languages as input. Here, the models are able to generate non-trivial summaries for some language pairs but still lag behind fully supervised models by a significant margin. We have included Figures 10 and 11 in the Appendix to illustrate this.

Furthermore, we ran inference with the m2m model on distant low-resource language pairs that were absent in training. Their LaSE scores were substantially below supervised pairs, meaning zero-shot transfer in supervised multilingual models (Johnson et al., 2017) shows weak performance.

We do not perform few-shot experiments and leave them as potential future directions.

6 Analysis of Results

Statistical significance While the scores obtained from the experiments in Section 5 indicate that the proposed m2m model performs better than the others, the differences are very close in many language pairs. Therefore, a statistical significance test is still warranted to support our claim further. As such, for each language pair experimented on, we performed the Bootstrap resampling test (Koehn, 2004) with the m2m model against the best-performing model among the others in a one vs. all manner: if m2m has the best (ROUGE-2/LaSE) score, we compare it with the model with

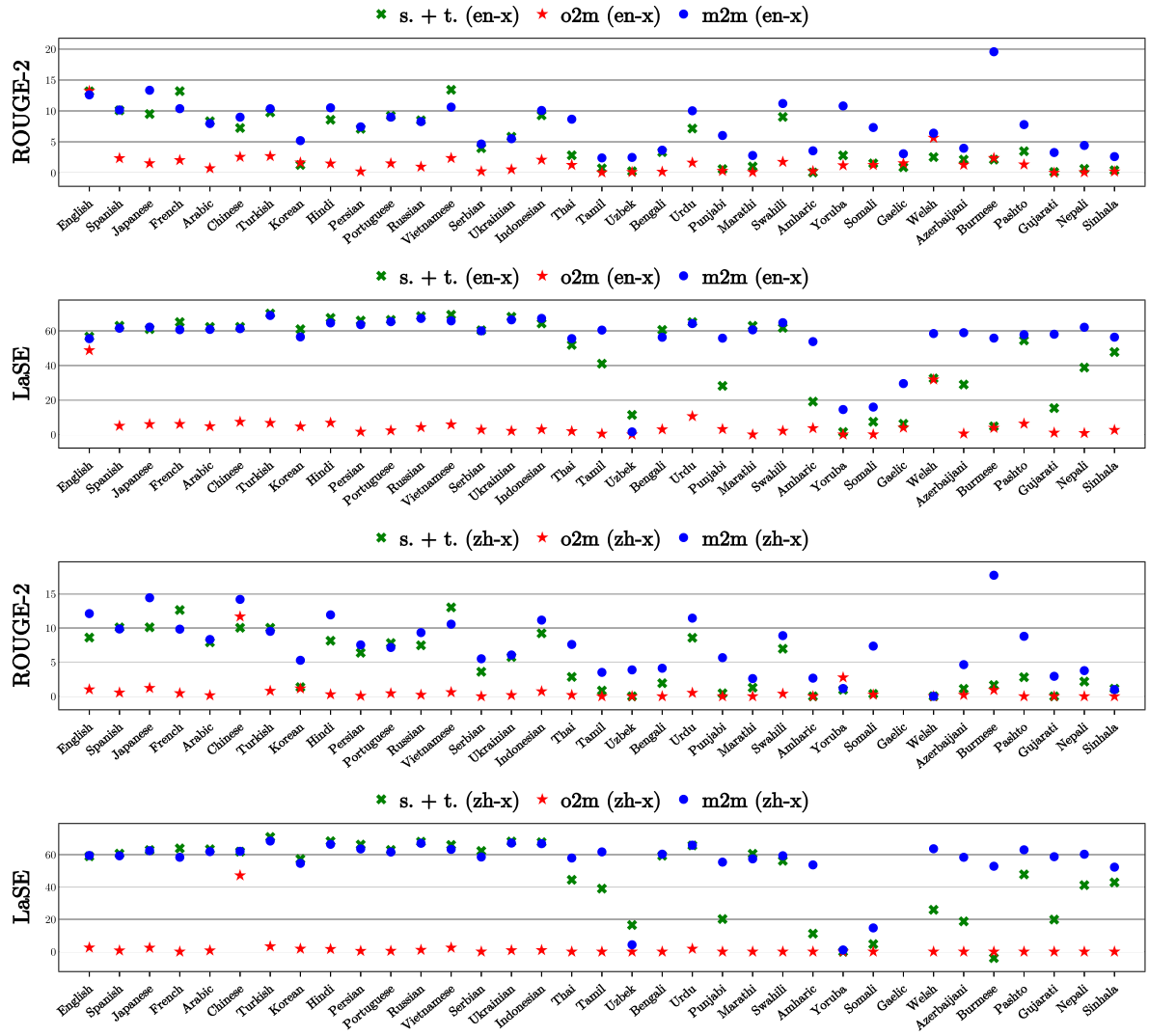


Figure 5: ROUGE-2 and LaSE scores for English and Chinese as source languages as the target languages vary. The m2m model significantly outperforms the o2m models and summarize-then-translate baseline in most languages. The comparisons with other source languages are shown in the Appendix (Figure 9) due to space limitations.

the second-best score, and if m2m is not the best, we compare it with the best.

Pivot	Metric	Better	Worse	Insignificant
x-en	R-2/LaSE	8/18	2/2	25/15
en-x	R-2/LaSE	20/15	3/14	12/6
x-zh	R-2/LaSE	11/13	0/0	23/21
zh-x	R-2/LaSE	17/12	1/2	16/20
x-hi	R-2/LaSE	18/15	1/6	15/13
hi-x	R-2/LaSE	19/15	0/6	15/13
x-ar	R-2/LaSE	6/15	2/3	26/16
ar-x	R-2/LaSE	23/15	1/5	10/14
x-ru	R-2/LaSE	6/11	2/7	26/16
ru-x	R-2/LaSE	19/13	2/7	13/14

Table 1: Significance test on different pivot languages.

Results ($p < 0.05$) in Table 1 reveal that in more than 42% language pairs tested, m2m is significantly better, and in less than 10% pairs, it is considerably worse.[10] This provides additional evidence in support of our claim that the m2m model performs better than others.

How reliable is LaSE? At first, we validated the reliability of LaSE by showing its correlation with ROUGE-2. We took different checkpoints of the in-language summarization model used in s.+t. and computed ROUGE-2 and LaSE for the nine languages in Section 3 for each checkpoint. The correlation coefficients of the calculated scores are shown in the second column of Table 2. For all languages (from high- to low-resource), LaSE has

[10]The numbers are even better if compared one vs. one.

a near-perfect correlation with ROUGE-2.

However, the purpose of LaSE is to show that it is language-agnostic and can even be computed in the absence of references in the target language. Therefore, we evaluate the summaries with references in a different language from the target using the m2m model. For each target language, we first compute the standard LaSE for different source languages (denoted as LaSE-in-lang). We again compute LaSE after swapping the reference texts with the references in the language of the input text[11] (denoted as LaSE-out-lang). We then show the correlation between the two variants of LaSE in the third column of Table 2[12] for each target language. Results show a substantial correlation between the two variants of LaSE for all languages.

From these two experiments, we can conclude that LaSE is an ideal metric for the evaluation of summarization systems and can be computed in a language-independent manner.

Target Lang.	ROUGE-2 vs. LaSE-in-lang. Pearson/Spearman	LaSE-in-lang vs. LaSE-out-lang. Pearson/Spearman
English	0.976/0.939	0.993/1.000
Arabic	0.903/0.987	0.968/0.942
Chinese	0.983/1.000	0.996/1.000
Indonesian	0.992/0.975	0.872/0.828
Bengali	0.947/0.902	0.819/0.771
Urdu	0.997/0.951	0.774/0.828
Punjabi	0.988/0.963	0.881/0.885
Swahili	0.990/0.951	0.979/0.885
Pashto	0.994/0.987	0.883/0.885

Table 2: Correlation analysis of ROUGE-2 and LaSE. We compute both Pearson and Spearman coefficients.

7 Related Works

Pipeline-based methods were popular at the beginning stages of XLS research (Leuski et al., 2003; Orasan and Chiorean, 2008; Wan et al., 2010), breaking the task into a sequence of summarization and translation tasks. End-to-end methods that performed XLS with a single model gained popularity with the emergence of neural models. Ayana et al. (2018) used knowledge distillation (Hinton et al., 2015) to train a student XLS model from two summarization and translation teacher models. Using a synthetic dataset, Zhu et al. (2019); Cao et al. (2020a) performed XLS with a dual Transformer (Vaswani et al., 2017) architecture in a multitask framework, while Bai et al. (2021) proposed a single encoder-decoder for better transfer across tasks. Chi et al. (2021) introduced multiple pretraining objectives specifically tailored to cross-lingual tasks that showed improved results on XLS. We refer our readers to Wang et al. (2022) for a more comprehensive literature review.

Until recently, XLS was limited primarily to English-Chinese due to the lack of benchmark datasets. To promote the task beyond this language pair, Ladhak et al. (2020) introduced Wikilingua, a large-scale many-to-one dataset with English as the pivot language, while Perez-Beltrachini and Lapata (2021) introduced XWikis, containing 4 languages in 12 directions.

More recently, Wang et al. (2023) explored zero-shot cross-lingual summarization by prompting (Liu et al., 2023) large language models like ChatGPT[13], GPT-4 (OpenAI, 2023), and BLOOMZ (Muennighoff et al., 2022).

8 Conclusion & Future Works

In this work, we presented CrossSum, a large-scale, non-English-centric XLS dataset containing 1.68 million samples in 1,500+ language pairs. CrossSum provides the first publicly available XLS dataset for many of these pairs. Performing a limited-scale human evaluation of CrossSum, we introduced MLS, a multistage sampling algorithm for general-purpose cross-lingual generation, and LaSE, a language-agnostic metric for evaluating summaries when reference summaries in the target languages may not be available. We demonstrated that training one multilingual model can help towards better XLS than baselines. We also shed light on the potential to perform zero-shot and few-shot XLS with CrossSum. We share our findings and resources in the hopes of making the XLS research community more inclusive and diverse.

In the future, we will investigate the use of CrossSum for other summarization tasks, e.g., multi-document (Fabbri et al., 2019) and multi-modal summarization (Zhu et al., 2018). We would also like to explore better techniques for m2m, zero-shot, and few-shot cross-lingual summarization.

[11] Our curation method ensures that such summaries always exist in the corresponding test sets.

[12] Since many test sets of the language pairs from Section 3 have too few samples for reliable evaluation (e.g., Punjabi-Pashto), for each target language, we use only the top-5 source languages by the number of their test set samples.

[13] https://openai.com/blog/chatgpt

Limitations

Though we believe that our work has many merits, some of its limitations must be acknowledged. Despite exhaustive human annotation being the most reliable means of ensuring the maximum quality of a dataset, we had to resort to the automatic curation of CrossSum due to the enormous scale of the dataset. As identified in the human evaluation, not all of the alignments made by LaBSE are correct. They are primarily summaries describing similar (i.e., having a substantial degree of syntactic or semantic similarity) but non-identical events. LaBSE also fails to penalize numerical mismatches, especially if the summaries depict the same event.

Consequently, any mistake made by LaBSE in the curation phase may propagate to the models trained using CrossSum. And since LaBSE is a component of the proposed LaSE metric, these biases may remain unidentified by LaSE in the evaluation stage. However, no matter which automatic method we use, there will be such frailties in these extreme cases. Since the objective of this paper is not to scrutinize the pitfalls of LaBSE but rather to use it as a means of curation and evaluation, we deem LaBSE the best choice due to its extensive language coverage and empirical performance in cross-lingual mining among existing alternatives.

Ethical Considerations

License CrossSum is a derivative of the XL-Sum dataset. XL-Sum has been released under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License (CC BY-NC-SA 4.0), allowing modifications and distributions for non-commercial research purposes. We are adhering to the terms of the license and releasing CrossSum under the same license.

Generated Text All of our models use the mT5 model as the backbone, which is pretrained on a large multilingual text corpus. For a text generation model, even small amounts of offensive or harmful texts in pretraining could lead to dangerous biases in generated text (Luccioni and Viviano, 2021). Therefore, our models can potentially generate offensive or biased content learned during the pretraining phase, which is beyond our control. Text summarization systems have also been shown to generate unfaithful and factually incorrect (albeit fluent) (Maynez et al., 2020) texts. Thus, we suggest carefully examining the potential biases before considering them in any real-world deployment.

Human Evaluation Annotators were hired from the graduates of an institute that provides professional training for many languages, including the ones evaluated in Section 3. Each annotator was given around 200-250 sequence pairs to evaluate. Each annotation took an average of one and a half minutes, with a total of approximately 5-6 hours for annotating the whole set. Annotators were paid hourly per the standard remuneration of bilingual professionals in local currency.

Environmental Impact A total of 25 models were trained as part of this work. Each model was trained for about three days on a 4-GPU Tesla P100 server. Assuming 0.08 kg/kWh carbon emission[14], less than 175kg of carbon was released into the environment in this work, which is orders of magnitude below the most computationally demanding models.

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[14] https://blog.google/technology/ai/minimizing-carbon-footprint/

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Appendix

A Aligning Summaries using LaBSE

In Section 2, we curated CrossSum by aligning parallel summaries in different languages. It might be argued why the articles themselves were not used for the alignment process. Initially, we experimented with whole-article embeddings. However, this resulted in many false-negative alignments, where similarity scores between parallel articles across languages were relatively low (verified manually between English and the authors' native languages). This is most likely attributed to the 512-token limit of LaBSE and different sequence lengths of those articles due to different languages having different subword segmentation fertility (Ács, 2019). This would entail that parallel articles in different languages might be truncated at different locations, resulting in discrepancies between their embeddings. As observed in the BUCC evaluation, LaBSE is well-suited for sentence-level retrieval. Since summaries are good representatives of entire articles, we finally chose summaries as our candidates for the alignment.

B Inter-annotator Agreement of Human Evaluation

Language Pair	Cohen's Kappa
Arabic-English	0.82
Chinese-English	0.73
Indonesian-English	0.73
Bengali-English	0.73
Urdu-English	0.76
Punjabi-English	0.71
Swahili-English	0.78
Pashto-English	0.75

Table 3: Language pair-wise kappa scores.

C Modeling Details

C.1 Choice of Pretrained Model

Many pretrained multilingual text-to-text models are currently available, e.g., mBART (Liu et al., 2020), CRISS (Tran et al., 2020), MARGE (Lewis et al., 2020), and mT5 (Xue et al., 2021b). While mBART and mT5 are pretrained with multilingual objectives, CRISS and MARGE are pretrained with a cross-lingual one, which better suits our use case. However, we choose mT5 for fine-tuning because of its broad coverage of 101 languages with support for 41 of the 45 languages from CrossSum, in contrast to only 15 languages in mBART or CRISS and 26 in MARGE.

C.2 Summarize-then-translate (s. + t.)

The primary reason for using summarize-then-translate rather than translate-then-summarize is the computational cost between these two. Available translation models only work for short sequences and are unsuitable for long documents. One solution is to segment the documents into sentences and then translate them. But that increases the compute overhead, and translations suffer from loss of context. We use a multilingual summarization model (Hasan et al., 2021) coupled with the multilingual machine translation model, M2M-100 (Fan et al., 2021), for our pipeline.

C.2.1 Multilingual Summarization

The pipeline first performs in-language summarization. We train our own model for summarization as the model released by Hasan et al. (2021) has been rendered unusable due to the change in the dataset split. We extend our component graphs to curate the in-language dataset splits. We consider articles having no parallel counterpart in any other language as single node components in the component graph. As before, we assign all articles originating from a single component to the training (dev/test) set of the dataset, extending them to the in-language splits too. We then train the multilingual model by fine-tuning mT5 with the in-language splits, sampling each batch of 256 samples from a single language with a sampling factor of $\alpha = 0.5$.

C.2.2 Multilingual Translation

For multilingual translation, we used M2M-100 (Fan et al., 2021) (418M parameters variant), a many-to-many multilingual translation model, with support for 37 languages from CrossSum.

C.3 Many-to-One (m2o) Model

Many-to-one training is standard for evaluating cross-lingual summarization. In these models, the language of the source text can vary, but the target language remains the same, i.e., as the pivot language. Instead of sampling all samples of a batch from the same language pair, we sample 8 mini-batches of 32 samples using a sampling factor of $\alpha = 0.25$, the source side of each originating from

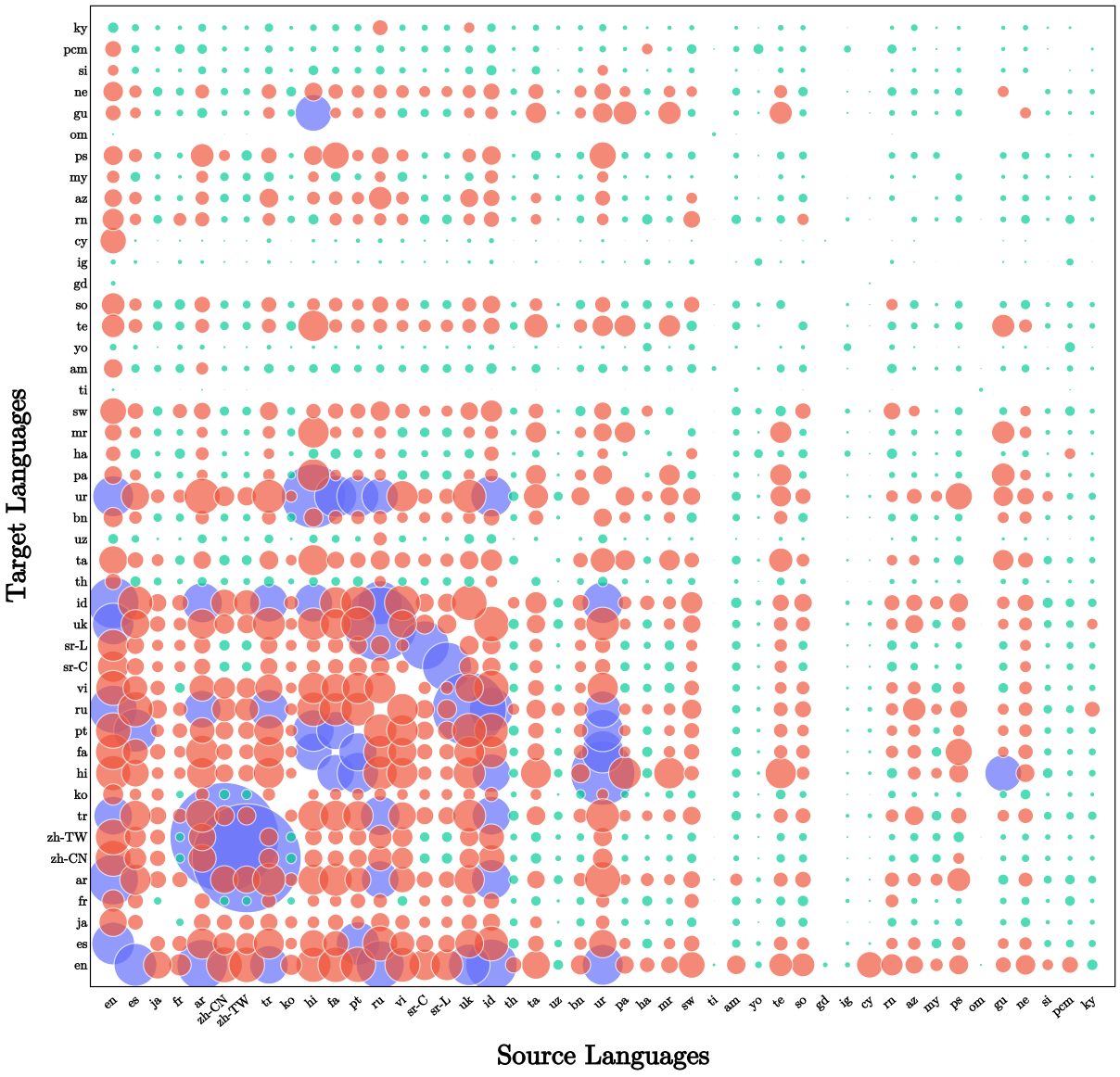


Figure 6: A bubble plot depicting the article-summary frequencies of CrossSum. The radii of the bubbles are proportional to the number of samples for the corresponding language pair (exact numbers are in Table 4). Languages are ordered by the language taxonomy from Joshi et al. (2020). To show better contrast between language pairs, we color a bubble cyan if its frequency is below 500 (1218 pairs), red for 500 to 5000 (688 pairs), and blue for frequencies exceeding 5000 (52 pairs).

a single language while the target language remains fixed. We then merge the mini-batches into a single batch and update the model parameters. This is to ensure that there are not many duplicates in a single batch (if all 256 samples of a batch are sampled from a single language pair, there might be many duplicates as many language pairs do not have 256 training samples) and the model still benefits the advantages of low-resource upsampling.

C.4 One-to-many (o2m) Model

o2m models are complementary to m2o models: we train them by keeping the source language fixed and varying the target language. We upsample the low-resource target languages with the same sampling factor of $\alpha = 0.25$ and merge 8 mini-batches of 32 samples each, analogous to m2o models.

C.5 Many-to-many (m2m) Multistage Model

This is the model obtained from the Algorithm 1. In contrast to standard language sampling (Conneau

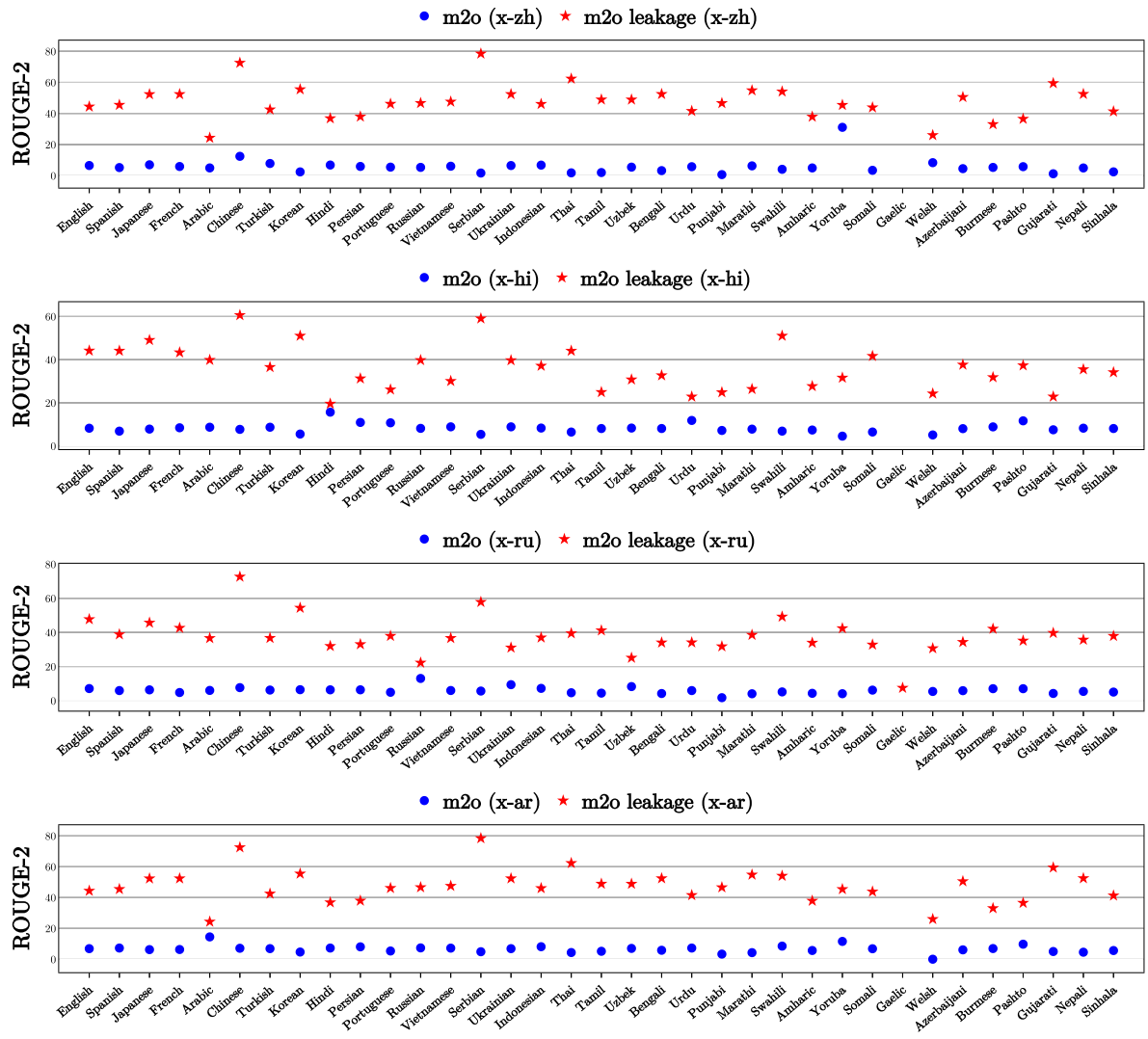


Figure 7: Training on the dataset respecting the original XL-Sum splits causes absurdly high ROUGE scores (marked red) in many-to-one models due to implicit data leakage. Therefore, we split taking the issue into account, and consequently, models trained on the new set (marked blue) do not exhibit any unusual spike in ROUGE-2.

et al., 2020), we sample the target language and then choose the source based on that decision. We use batch size 256, 8 mini-batches with size 32, and $\alpha = 0.5, \beta = 0.75$.

C.6 Many-to-many (m2m) Unistage Model

This algorithm is similar to standard language sampling, the difference being that languages are sampled as pairs from all possible combinations. Instead of sampling one language pair at each training step, we sample 8 pairs, one for each mini-batch of size 32. We then merge the mini-batches into a single batch of 256 samples before updating the model parameters. We use a sampling factor of $\alpha = 0.25$.

In all models, we discarded a language pair from training if it had fewer than 30 training samples to prevent too many duplicates in a mini-batch. The training was done together with the in-language samples.

D Experimental Details

D.1 Training Setups

Fine-tuning generation models is compute-intensive, and due to computational limitations, we fine-tune all pretrained models for 25k steps with an effective batch size of 256, which roughly takes about three days on a 4-GPU NVIDIA P100 server. We use the base variant of mT5, having 250k vocabulary, 768 embedding and dimension size, 12 attention heads, and 2048 FFN size, with 580M parameters. We limit the input to 512 and output to 84 tokens. All models are trained on the

respective subsets of the CrossSum training set.

D.2 Inference

During inference, we jump-start the decoder with language-specific `BOS` (beginning of sequence) tokens (Johnson et al., 2017) at the first decoding step for guiding the decoder to generate summaries in the intended target language. We use beam search (Medress et al., 1977) with the beam size 4 and use a length penalty (Wu et al., 2016) of 0.6.

E Ablation Studies

We make several design choices in the multistage sampling algorithm. We break them into two main decisions:

1. Making mini-batches and sampling the language pair for each mini-batch.

2. Keeping either the source or the target language fixed for each batch.

To verify that these choices indeed affect performance positively, we train five different models for ablation:

1. Sampling the language pair in mini-batches in one stage only and then merging them into large batches before updating model parameters: m2m-unistage.

2. Sampling the language pair with large batches of 256 samples without mini-batching: m2m-large.

3. Multistage sampling keeping only the target language fixed in a batch: m2m-tgt *[our proposed model]*.

4. Multistage sampling keeping only the source language fixed in a batch: m2m-src; i.e., the complement of our proposed model.

5. Multistage sampling keeping either the source or the target language fixed (with equal probability) for each batch: m2m-src-tgt.

We benchmark on all the language pairs done previously and show the mean ROUGE-2 and LaSE scores in Table 5.

Model	Scores	Significance		
	R-2/LaSE	Better	Worse	Insignificant
m2m-large	**8.31/57.45**	122	**59**	503
m2m-unistage	7.51/55.36	191	149	344
m2m-tgt	8.15/57.15	**289**	66	**329**
m2m-src	4.44/26.75	34	477	173
m2m-src-tgt	6.47/42.55	89	297	298

Table 5: ROUGE-2 and LaSE scores for ablation.

As can be seen from the table, m2m-large, the standard m2m model, has the best average ROUGE-2/LaSE scores among all m2m variants. This begs the question of whether our proposed multistage sampling is, after all, needed or not. But the scores of the proposed m2m-tgt model do not fall much below. Therefore, we show statistical significance test results of all m2m models, comparing them against m2o, o2m, and s.+t. in one vs. all manner.

Significance results paint a different picture: m2m-tgt triumphs over all other models, getting significantly better results on 42% language pairs, more than double the m2m-large model. We inspected the results individually and found that the results are notably better on language pairs that are not adequately represented in the training set. m2m-tgt performs comparatively worse on high-resource language pairs, which we think is a fair compromise to uplift low-resource ones. As m2m-large can sample a pair only once per batch, it fails to incorporate many language pairs due to them having insufficient participation during training. On the other hand, our proposed multistage sampling algorithm performs well in this regard by sampling in two stages.

While m2m-tgt outperforms all the rest, m2m-src falls behind all other models by a large margin. This phenomenon also has the same trend as the results in Section 5, where o2m models failed at generating cross-lingual summaries. This is also in line with our hypothesis made, as m2m-src and m2m-tgt mimic the training settings of the o2m and m2o models, respectively, at the batch level. The m2m-src-tgt is the middle ground between m2m-src and m2m-tgt and, likewise, scores between these two. In our opinion, the performance dynamics between the m2o (m2m-tgt) and o2m (m2m-src) models is an interesting finding and should be studied in depth as a new research direction in future works.

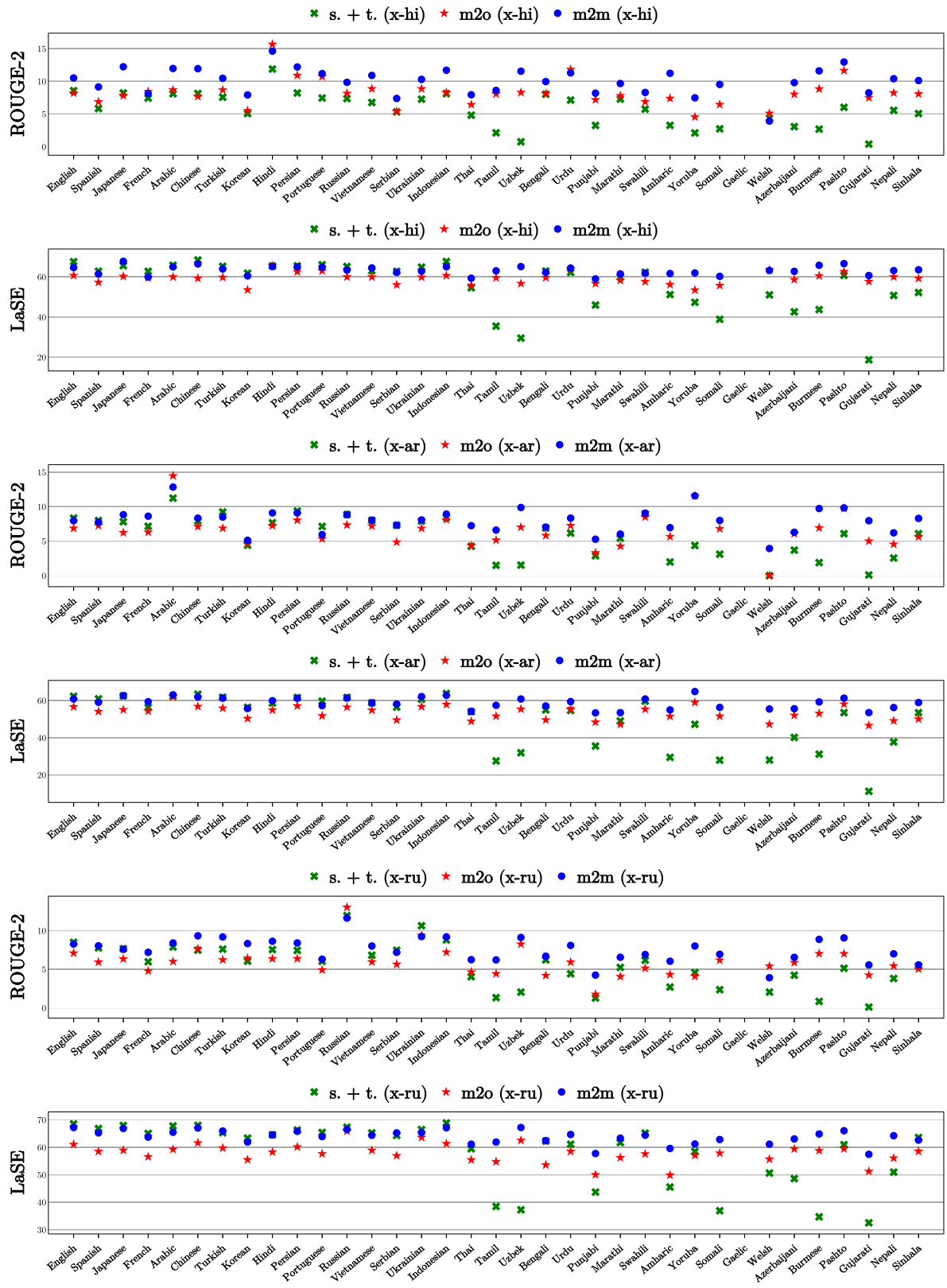


Figure 8: ROUGE-2 and LaSE scores for Hindi, Arabic, and Russian as target pivots as the sources languages vary. Just like Figure 4, the m2m model significantly outperforms the m2o models and s. + t. baseline on most languages.



Figure 9: ROUGE-2 and LaSE scores for Hindi, Arabic, and Russian as source pivots as the target languages vary. Just like Figure 5, the m2m model significantly outperforms the o2m models and s. + t. baseline on most languages.

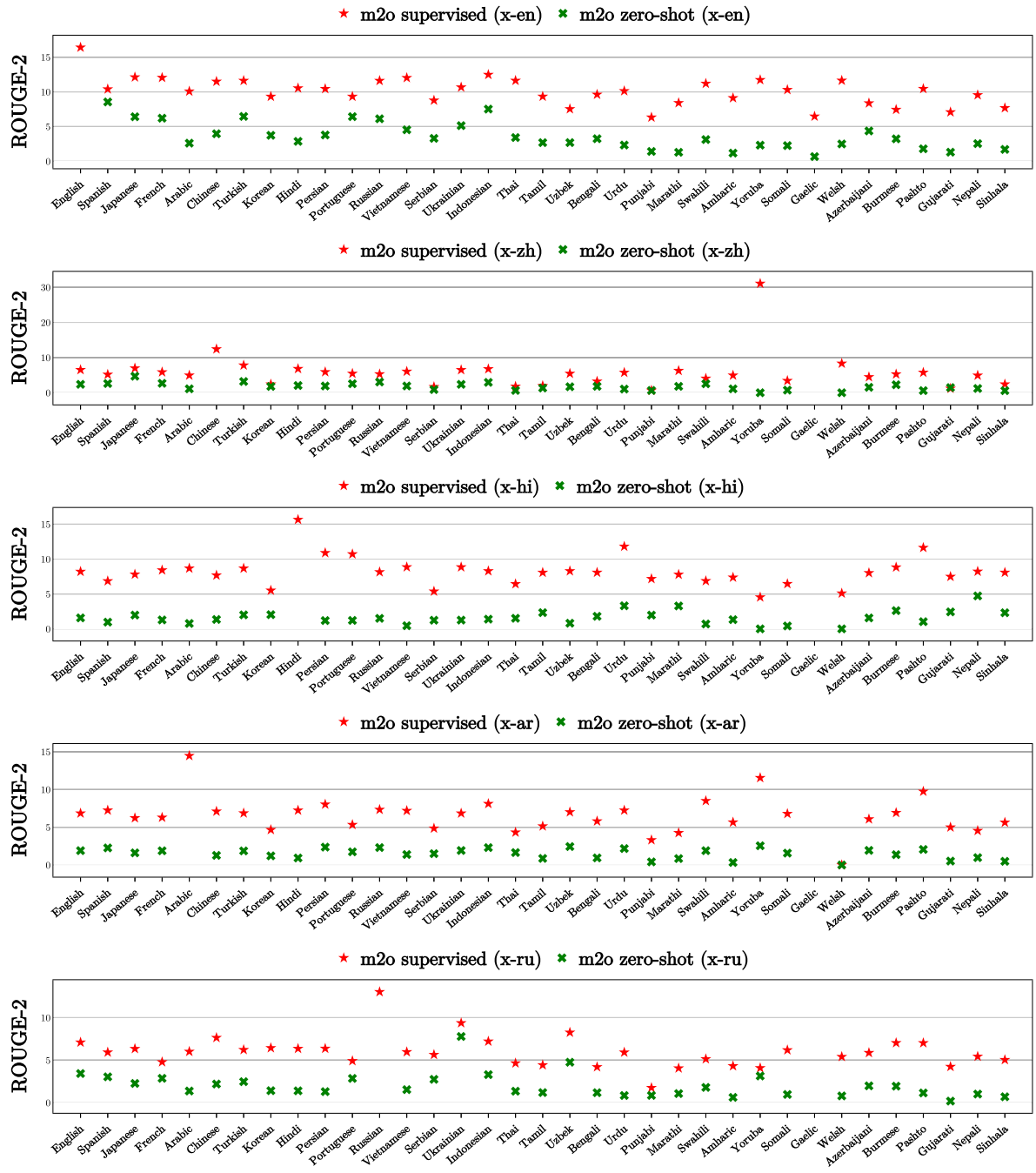


Figure 10: Zero-shot ROUGE-2 scores for the different target languages as the source languages vary. The zero-shot models are trained with only the in-language samples of the pivot. Though their results are clearly behind the fully supervised models, the zero-shot models are able to generate non-trivial summaries for many language pairs.

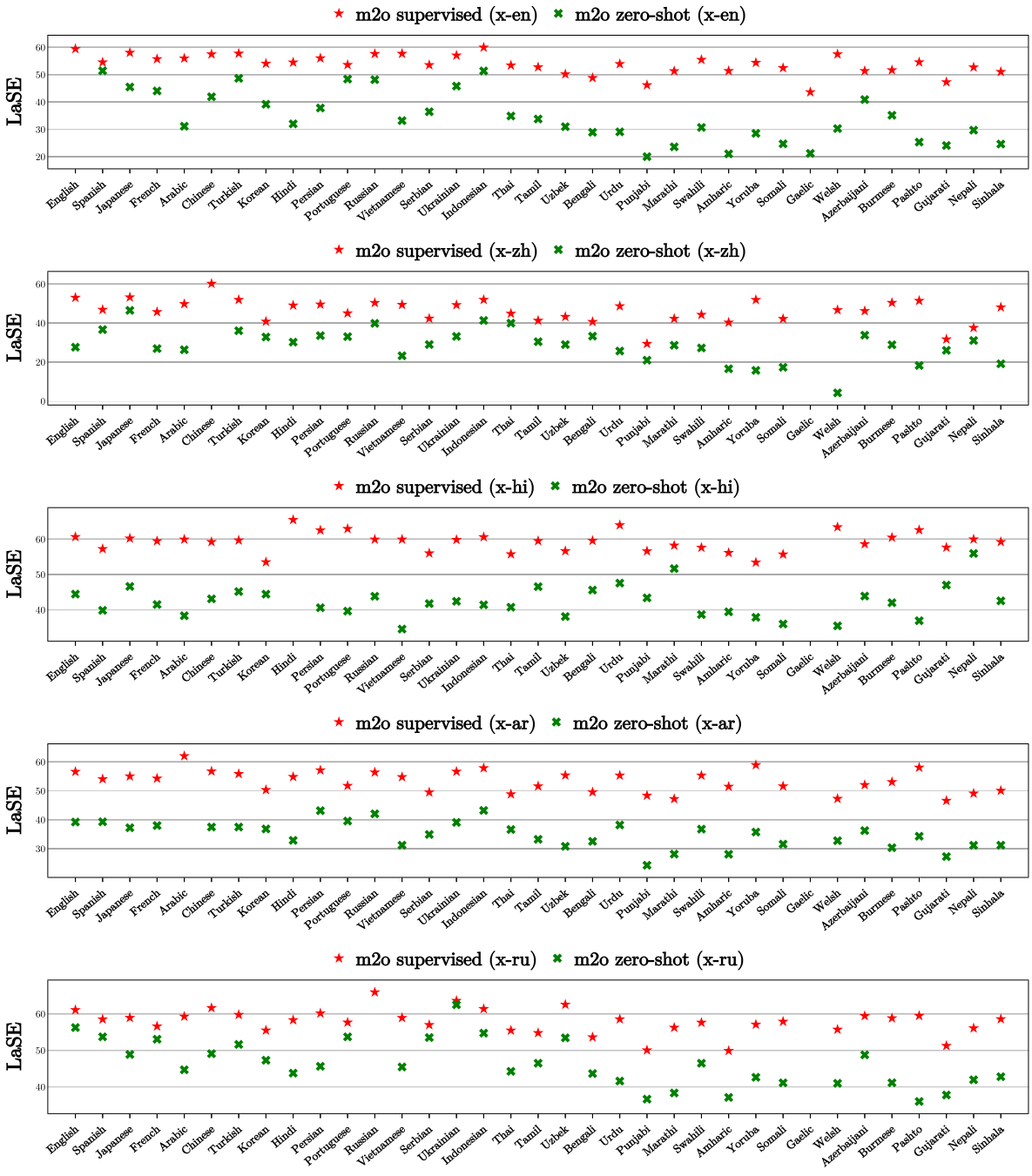


Figure 11: Zero-shot LaSE scores for the different source languages as the target languages vary. The zero-shot models are trained with only the in-language samples of the pivot. Though their results are clearly behind the fully supervised models, the zero-shot models are able to generate non-trivial summaries for many language pairs.

Language	am	ar	az	bn	my	zh-CN	zh-TW	en	fr	gu	ha	hi	ig	id	ja	rn	ko	ky	mr	ne	om	ps	fa	pcm	pt	pa	ru	gd	sr-C	sr-L	si	so	es	sw	ta	te	th	ti	tr	uk	ur	uz	vi	cy	yo	Total
am	–	659	95	274	95	179	169	1445	371	171	220	361	31	497	269	415	239	93	223	304	19	189	423	205	291	191	333	0	350	361	62	299	346	383	374	322	122	129	424	341	393	40	287	1	71	**12066**
ar	659	–	781	799	646	2905	2783	9630	991	467	733	3651	83	6061	1175	873	691	302	547	844	9	2148	4170	427	2507	541	5329	1	1101	1139	316	1049	3650	1175	1294	852	371	29	4106	3429	4900	381	2623	39	141	**76348**
az	95	781	–	283	81	363	324	1307	203	181	124	735	26	1111	226	178	162	228	198	246	2	249	814	93	668	186	2087	3	286	285	124	359	704	535	505	233	139	2	1476	1373	957	195	726	31	40	**18924**
bn	274	799	283	–	145	308	275	1544	320	551	231	1376	37	1072	344	297	351	154	580	665	2	296	787	132	769	574	792	0	559	560	154	411	697	477	913	783	245	6	857	692	1381	96	521	35	62	**21407**
my	95	646	81	145	–	349	321	694	88	99	71	522	10	767	148	105	116	53	91	147	1	237	432	38	232	86	528	1	117	120	88	79	438	81	180	147	73	4	442	356	580	62	450	2	11	**9333**
zh-CN	179	2905	363	308	349	–	44561	4864	329	197	151	1331	34	2787	1010	227	407	135	236	269	13	552	1091	144	1334	235	2396	2	467	496	167	330	1941	402	500	352	263	13	1482	1591	1613	171	1853	28	40	**78118**
zh-TW	169	2783	324	275	321	44561	–	4777	307	167	135	1167	31	2573	955	208	384	125	205	248	15	499	947	134	1224	219	2166	1	418	457	160	302	1817	372	455	328	243	15	1273	1438	1420	162	1655	26	39	**75500**
en	1445	9630	1307	1544	694	4864	4777	–	1891	973	916	4668	147	10012	3035	1870	1686	497	1172	1600	35	1514	4717	1076	4714	1315	8680	127	3748	3798	525	2139	6891	2701	3134	2111	1014	58	5612	6530	6319	450	4580	2636	229	**127381**
fr	371	991	203	320	88	329	307	1891	–	227	476	607	105	1020	275	723	270	118	238	322	5	189	609	440	913	237	802	2	553	570	102	499	987	870	423	379	180	12	820	717	767	73	442	40	163	**19675**
gu	171	467	181	551	99	197	167	973	227	–	138	5087	37	706	217	180	263	101	2057	547	1	238	511	98	524	2161	550	1	337	339	132	256	532	307	1728	2020	162	5	616	506	1605	69	442	23	49	**25578**
ha	220	733	124	231	71	151	135	916	476	138	–	454	202	897	163	484	141	61	155	238	6	222	480	518	372	145	507	1	248	259	52	386	456	566	294	250	85	8	511	405	522	56	357	31	361	**13088**
hi	361	3651	735	1376	522	1331	1167	4668	607	5087	454	–	60	5598	619	479	509	231	3757	1340	3	1504	5293	187	6478	3971	4434	2	806	808	442	732	2917	896	3631	3696	367	9	3667	3912	15502	342	3706	80	77	**96014**
ig	31	83	26	37	10	34	31	147	105	37	202	60	–	116	23	105	28	17	52	40	5	9	48	251	62	39	79	0	45	48	12	72	87	151	56	50	16	5	92	74	60	11	61	6	291	**2814**
id	497	6061	1111	1072	767	2787	2573	10012	1020	706	897	5598	116	–	1271	986	784	348	755	1101	9	1450	3883	363	4375	718	7274	5	1377	1373	478	1303	4540	1873	1867	1129	603	11	5630	4799	6468	428	4790	146	172	**93526**
ja	269	1175	226	344	148	1010	955	3035	275	217	163	619	23	1271	–	368	660	143	298	417	3	270	1014	154	701	264	1419	2	555	568	112	388	950	426	631	420	307	4	1242	1016	806	54	901	22	31	**23876**
rn	415	873	178	297	105	227	208	1870	723	180	484	479	105	986	368	–	279	108	237	370	17	227	677	392	510	196	670	0	442	441	80	580	595	1183	507	351	146	13	709	609	614	55	613	19	173	**18311**
ko	239	691	162	351	116	407	384	1686	270	263	141	509	28	784	660	279	–	94	314	448	1	149	582	136	581	269	617	1	522	536	87	240	607	318	530	441	190	4	672	611	527	54	524	15	46	**16086**
ky	93	302	228	154	53	135	125	497	118	101	61	231	17	348	143	108	94	–	105	155	4	97	251	60	247	117	955	1	200	207	50	151	259	145	205	175	111	4	340	505	263	113	208	9	26	**7771**
mr	223	547	198	580	91	236	205	1172	238	2057	155	3757	52	755	298	237	314	105	–	617	2	228	604	137	532	1759	633	1	422	440	131	263	593	327	1746	1870	194	10	704	590	1381	75	473	15	50	**25017**
ne	304	844	246	665	147	269	248	1600	322	547	238	1340	40	1101	417	370	448	155	617	–	1	291	915	127	703	530	815	2	547	545	164	410	681	511	973	741	227	7	923	744	1154	81	714	31	66	**21821**
om	19	9	2	2	1	13	15	35	5	1	6	3	5	9	3	17	1	4	2	1	–	2	4	10	4	3	8	0	4	6	0	6	9	4	3	2	2	100	4	11	1	4	2	1	5	**348**
ps	189	2148	249	296	237	552	499	1514	189	238	222	1504	9	1450	270	227	149	97	228	291	2	–	2788	92	591	250	1213	0	220	231	146	305	763	314	435	308	90	7	1033	818	2812	160	657	7	33	**23833**
fa	423	4170	814	787	432	1091	947	4717	609	511	480	5293	48	3883	1014	677	582	251	604	915	4	2788	–	191	5461	523	4125	1	1011	1011	265	820	2532	1002	1223	775	363	8	3644	3542	6694	306	3167	68	73	**67845**
pcm	205	427	93	132	38	144	134	1076	440	98	518	187	251	363	154	392	136	60	137	127	10	92	191	–	229	106	306	0	240	247	30	220	315	428	219	154	88	26	279	284	227	19	174	7	462	**9465**
pt	291	2507	668	769	232	1334	1224	4714	913	524	372	6478	62	4375	701	510	581	247	532	703	4	591	5461	229	–	553	4247	7	1359	1343	232	612	7071	984	1034	806	472	4	3451	4374	6654	182	3732	110	96	**71345**
pa	191	541	186	574	86	235	219	1315	237	2161	145	3971	39	718	264	196	269	117	1759	530	3	250	523	106	553	–	589	2	399	399	126	288	566	356	1667	1854	195	11	615	562	1484	68	425	12	39	**24845**
ru	333	5329	2087	792	528	2396	2166	8680	802	550	507	4434	79	7274	1419	670	617	955	633	815	8	1213	4125	306	4247	589	–	4	1427	1413	354	1097	4652	1557	1526	849	557	9	5906	20706	5036	765	3759	131	115	**101417**
gd	0	1	3	0	1	2	1	127	2	1	1	2	0	5	2	0	1	1	1	2	0	0	1	0	7	2	4	–	2	3	2	1	3	1	1	1	1	0	6	4	3	0	4	36	2	**237**
sr-C	350	1101	286	559	117	467	418	3748	553	337	248	806	45	1377	555	442	522	200	422	547	4	220	1011	240	1359	399	1427	2	–	9000	124	375	1225	564	748	677	337	6	1248	1514	1013	109	674	43	72	**35491**
sr-L	361	1139	285	560	120	496	457	3798	570	339	259	808	48	1373	568	441	536	207	440	545	6	231	1011	247	1343	399	1413	3	9000	–	133	381	1258	560	768	688	345	9	1239	1506	1009	109	631	45	74	**35758**
si	62	316	124	154	88	167	160	525	102	132	52	442	12	478	112	80	87	50	131	164	0	146	265	30	232	126	354	2	124	133	–	132	259	186	345	172	71	6	302	309	512	39	217	8	14	**7422**
so	299	1049	359	411	79	330	302	2139	499	256	386	732	72	1303	388	580	240	151	263	410	6	305	820	220	612	288	1097	1	375	381	132	–	682	1024	712	373	172	17	955	874	1005	73	729	21	110	**21232**
es	346	3650	704	697	438	1941	1817	6891	987	532	456	2917	87	4540	950	595	607	259	593	681	9	763	2532	315	7071	566	4652	3	1225	1258	259	682	–	1045	1051	831	480	12	3617	3119	3046	287	2318	55	134	**65018**
sw	383	1175	535	477	81	402	372	2701	870	307	566	896	151	1873	426	1183	318	145	327	511	4	314	1002	428	984	356	1557	1	564	560	186	1024	1045	–	934	495	264	11	1350	1294	1243	81	928	35	216	**28575**
ta	374	1294	505	913	180	500	455	3134	423	1728	294	3631	56	1867	631	507	530	205	1746	973	3	435	1223	219	1034	1667	1526	1	748	768	345	712	1051	934	–	2236	388	12	1467	1414	2393	114	1069	32	72	**39809**
te	322	852	233	783	147	352	328	2111	379	2020	250	3696	50	1129	420	351	441	175	1870	741	2	308	775	154	806	1854	849	1	677	688	172	373	831	495	2236	–	306	11	875	832	1743	99	634	20	62	**31453**
th	122	371	139	245	73	263	243	1014	180	162	85	367	16	603	307	146	190	111	194	227	2	90	363	88	472	195	557	1	337	345	71	172	480	264	388	306	–	3	469	482	424	33	355	13	23	**10991**
ti	129	29	2	6	4	13	15	58	12	5	8	9	5	11	4	13	4	4	10	7	100	7	8	26	4	11	9	0	6	9	6	17	12	11	12	11	3	–	9	9	5	2	4	0	6	**635**
tr	424	4106	1476	857	442	1482	1273	5612	820	616	511	3667	92	5630	1242	709	672	340	704	923	4	1033	3644	279	3451	615	5906	6	1248	1239	302	955	3617	1350	1467	875	469	9	–	4085	4314	361	2953	127	128	**70035**
uk	341	3429	1373	692	356	1591	1438	6530	717	506	405	3912	74	4799	1016	609	611	505	590	744	11	818	3542	284	4374	562	20706	4	1514	1506	309	874	3119	1294	1414	832	482	9	4085	–	4252	438	2992	105	92	**83856**
ur	393	4900	957	1381	580	1613	1420	6319	767	1605	522	15502	60	6468	806	614	527	263	1381	1154	1	2812	6694	227	6654	1484	5036	3	1013	1009	512	1005	3046	1243	2393	1743	424	5	4314	4252	–	391	3707	70	85	**95355**
uz	40	381	195	96	62	171	162	450	73	69	56	342	11	428	54	55	54	113	75	81	4	160	306	19	182	68	765	0	109	109	39	73	287	81	114	99	33	2	361	438	391	–	259	11	18	**6896**
vi	287	2623	726	521	450	1853	1655	4580	442	442	357	3706	61	4790	901	613	524	208	473	714	2	657	3167	174	3732	425	3759	4	674	631	217	729	2318	928	1069	634	355	4	2953	2992	3707	259	–	101	78	**55495**
cy	1	39	31	35	2	28	26	2636	40	23	31	80	6	146	22	19	15	9	15	31	1	7	68	7	110	12	131	36	43	45	8	21	55	35	32	20	13	0	127	105	70	11	101	–	8	**4301**
yo	71	141	40	62	11	40	39	229	163	49	361	77	291	172	31	173	46	26	50	66	5	33	73	462	96	39	115	2	72	74	14	110	134	216	72	62	23	6	128	92	85	18	78	8	–	**4155**

Table 4: An article-summary statistics of the CrossSum dataset containing a total of 1,678,466 cross-lingual samples. The rows indicate the articles' language, and the columns of their summaries'. For example, the cell on the second column of the fourth row indicates the number of samples where the article is in Bengali and the summary in Arabic.

ACL 2023 Responsible NLP Checklist

A For every submission:

☑ A1. Did you describe the limitations of your work?
In the Limitations section after the Conclusion & Future Works

☑ A2. Did you discuss any potential risks of your work?
In the Limitations and Ethical Considerations sections after the Conclusion & Future Works

☑ A3. Do the abstract and introduction summarize the paper's main claims?
Section 1

☒ A4. Have you used AI writing assistants when working on this paper?
Left blank.

B ☑ Did you use or create scientific artifacts?

Section 2

☑ B1. Did you cite the creators of artifacts you used?
Section 5 and Appendix C

☑ B2. Did you discuss the license or terms for use and / or distribution of any artifacts?
In the Ethical Considerations section after the Conclusion & Future Works

☑ B3. Did you discuss if your use of existing artifact(s) was consistent with their intended use, provided that it was specified? For the artifacts you create, do you specify intended use and whether that is compatible with the original access conditions (in particular, derivatives of data accessed for research purposes should not be used outside of research contexts)?
In the Ethical Considerations section after the Conclusion & Future Works

☒ B4. Did you discuss the steps taken to check whether the data that was collected / used contains any information that names or uniquely identifies individual people or offensive content, and the steps taken to protect / anonymize it?
The dataset is a derivative of a previous work that has already addressed the aforementioned issues.

☑ B5. Did you provide documentation of the artifacts, e.g., coverage of domains, languages, and linguistic phenomena, demographic groups represented, etc.?
Figure 6

☑ B6. Did you report relevant statistics like the number of examples, details of train / test / dev splits, etc. for the data that you used / created? Even for commonly-used benchmark datasets, include the number of examples in train / validation / test splits, as these provide necessary context for a reader to understand experimental results. For example, small differences in accuracy on large test sets may be significant, while on small test sets they may not be.
Table 4

C ☑ Did you run computational experiments?

Sections 5 and 6

☑ C1. Did you report the number of parameters in the models used, the total computational budget (e.g., GPU hours), and computing infrastructure used?
In the Ethical Considerations section after the Conclusion & Future Works, and Appendix C

The Responsible NLP Checklist used at ACL 2023 is adopted from NAACL 2022, with the addition of a question on AI writing assistance.

☑ C2. Did you discuss the experimental setup, including hyperparameter search and best-found hyperparameter values?
Section 5 and Appendix C

☑ C3. Did you report descriptive statistics about your results (e.g., error bars around results, summary statistics from sets of experiments), and is it transparent whether you are reporting the max, mean, etc. or just a single run?
Figures 4, 5, 8, 9, 10, and 11

☑ C4. If you used existing packages (e.g., for preprocessing, for normalization, or for evaluation), did you report the implementation, model, and parameter settings used (e.g., NLTK, Spacy, ROUGE, etc.)?
Section 5

D ☑ Did you use human annotators (e.g., crowdworkers) or research with human participants?
Section 3

☑ D1. Did you report the full text of instructions given to participants, including e.g., screenshots, disclaimers of any risks to participants or annotators, etc.?
Section 3

☑ D2. Did you report information about how you recruited (e.g., crowdsourcing platform, students) and paid participants, and discuss if such payment is adequate given the participants' demographic (e.g., country of residence)?
In the Ethical Considerations section after the Conclusion & Future Works

☑ D3. Did you discuss whether and how consent was obtained from people whose data you're using/curating? For example, if you collected data via crowdsourcing, did your instructions to crowdworkers explain how the data would be used?
In the Ethical Considerations section after the Conclusion & Future Works

☒ D4. Was the data collection protocol approved (or determined exempt) by an ethics review board?
Not applicable. ☐

☑ D5. Did you report the basic demographic and geographic characteristics of the annotator population that is the source of the data?
Section 3